

MANAGEMENT REVIEW

Management without theatre.

OPERATIONS

PROBLEM SOLVING WITH A3 AND PARETO

Why problems that are worked around come back, what the Pareto principle is and is not, how Toyota puts a problem on one sheet, and an A3 card to fill in.



239 h

of watching nurses in nine hospitals: most problems were worked around rather than fixed at the cause.

TUCKER & EDMONDSON, 2003

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Every operation has problems that come back every week: the same late start, the same missing part, the same complaint. This issue is about the difference between getting through the shift and making sure the problem does not return.

Researchers who followed nurses in nine hospitals found that most problems were worked around and few reached someone who could remove the cause. The Pareto chart shows where to start; Joseph Juran gave the idea Pareto's name and later admitted it was the wrong one. Toyota puts a whole problem, from the background to the follow-up, on one sheet of A3 paper. A study of 18 hospital projects suggests that following the steps matters.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

Worked around, not solved

Anita Tucker and Amy Edmondson followed nurses in nine hospitals as they met failures in their daily work, and recorded what they did next.

239 h

of observation on the wards

26

nurses followed through
their shifts

9

hospitals

First-order problem solving

The nurse finds a way around the problem

The patient is served, the shift goes on

The cause stays where it was

Second-order problem solving

The nurse deals with the immediate
problem

and tells someone who can change the
cause

The next shift may not meet it again

Working around was the dominant response. The nurses were not careless: under time pressure they got the job done, and in doing so kept the problem out of sight of the people who could remove it.

OUR READING

A workaround solves the problem for one shift. It also hides it from everyone who could make sure it does not come back.

Source: Anita L. Tucker & Amy C. Edmondson, California Management Review, 2003

THE PRINCIPLE

A principle with the wrong name

In the mid-1920s a young engineer, Joseph Juran, noticed that quality defects are not equal in frequency: a few kinds make up most of them. Years later he gave the idea the name of the Italian economist Vilfredo Pareto.

**1896**

Pareto publishes his work on the unequal distribution of incomes.

**1920s**

Juran sees that a few kinds of defects make up most of them.

**1974**

Juran's "Mea culpa": the name was a mistake. It stays anyway.

"I was forced to confess that I had mistakenly applied the wrong name to the principle."

That is how Juran put it in 1974. He had called it the principle of "the vital few and the trivial many". Pareto had written about wealth and incomes, not about defects; the step to quality was Juran's. Later Juran preferred "the vital few and the useful many", so that the rest would not be read as worthless.

OUR READING

80/20 is not a law of nature. It is a reminder to count before you decide where to start.

Source: Joseph M. Juran, Juran Institute, 1974

Juran's paper is dated 1974 in the copy of the Juran Institute; a bibliography lists it in Quality Progress in May 1975.

[More in the essay](#)[Pareto and 5 Why in practice, not in PowerPoint](#)

THE MODEL

One problem, one sheet

The A3 report is a practice Toyota pioneered: the problem, the analysis, the countermeasures and the plan on a single sheet of A3 paper, 297 × 420 mm. John Shook calls it standardised storytelling.

01	Background	Why this problem matters now.
02	Current condition	What happens today, drawn and measured.
03	Goal	What will be different, by how much, by when.
04	Analysis	Why it happens: the cause behind the cause.
05	Countermeasures	What will change, aimed at the cause, not the symptom.
06	Plan	Who does what, by when.
07	Follow-up	How the result is checked, and what happens if it does not come.

Shook writes that managers use the A3 to teach others to find root causes and think with evidence, and to bring people and departments to agreement.

OUR READING

The sheet is small on purpose. If the problem does not fit on one page, it has not been understood yet.

Sources: Lean Enterprise Institute, 2014; John Shook, MIT Sloan Management Review, 2009

The sections follow Shook and the Lean Enterprise Institute; the descriptions are the editors'.

HOW IT IS APPLIED

Eight steps and a check

In 2004 Toyota gathered its ways of solving problems into the Toyota Business Practices: eight steps, from describing the problem to making the solution the new standard.

Plan

1 Clarify the problem

2 Break it down

3 Set a target

4 Find the root cause

5 Develop countermeasures

Do, check, act

6 See the countermeasures through

7 Check the results and the process

8 Make what worked the standard

A study by Ghosh and Sobek looked at 18 improvement projects in one hospital. The projects that followed all the steps of the problem-solving routine improved more. It is one hospital and a correlation, not proof.

OUR READING

Many attempts fail at step one: the problem is written as a solution, “we need more people”, instead of as a gap between what should happen and what does.

Sources: Toyota Motor Corporation, 2012; M. Ghosh & D. K. Sobek II, Journal of Health Organization and Management, 2015

The steps are Toyota's, in our words; the grouping into plan, do, check and act follows summaries of the method.

HOW IT IS MEASURED

A Pareto chart in four steps

As ASQ describes it, the chart sorts the bars from the longest to the shortest and adds them up with a line.

01	Choose one measure	A count or a cost, over one period.
02	Group by cause	Categories you can act on; "other" goes last.
03	Sort and add up	Longest bar first, then the running percentage. Start on the left.
04	Draw it again	After the change, for the same period and the same categories.

REPORTED INJURIES TO EMPLOYEES BY KIND, GREAT BRITAIN, 2024/25



Sorted, the first two kinds make up 47% of the injuries employers reported, the first five 75%.
A safety A3 would start with the first two bars.

Sources: ASQ, 2026; Health and Safety Executive, 2025

Data: HSE, injuries employers must report (RIDDOR). The sums and the steps are the editors'.

[Open the tool: Pareto 80/20](#)

stivencatalyst.com/tools/pareto/

TOOL OF THE ISSUE

The A3 card

One sheet for one problem. Fill it in with the people who do the work, and in pencil: the sheet changes as you learn.

01 The problem and its owner in a few words, one name, a date

02 Background why it matters now

03 Current condition what happens today, with a number

04 Goal what changes, by how much, by when

05 Root cause why, and why again, until you reach something you can change

06 Countermeasures and plan what, who, by when

07 Follow-up when you check, and what you do if the number does not move

Source: Lean Enterprise Institute, 2014

A practice proposed by the editors. The sections follow the Lean Enterprise Institute and John Shook.

Open the tool: 5 Whys

stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Why Hospitals Don't Learn from Failures

Anita L. Tucker & Amy C. Edmondson,
California Management Review, 2003

<https://cmr.berkeley.edu/2003/02/45-2-why-hospitals-do-not-learn-from-failures-organizational-and-psychological-dynamics-that-inhibit-system-change>

The Non-Pareto Principle; Mea Culpa

Joseph M. Juran, Juran Institute, 1974

<https://www.juran.com/wp-content/uploads/2021/03/The-Non-Pareto-Principle-1974.pdf>

Lean Lexicon: A3 report

Lean Enterprise Institute, 2014

<https://www.lean.org/lexicon-terms/a3-report/>

Toyota's Secret: The A3 Report

John Shook, MIT Sloan Management Review, 2009

<https://sloanreview.mit.edu/article/toyotas-secret-the-a3-report/>

75 Years of Toyota: Toyota Business Practices

Toyota Motor Corporation, 2012

https://www.toyota-global.com/company/history_of_toyota/75years/text/leaping_forward_as_a_global_corporation/chapter4/section7/item4_a.html

A problem-solving routine for improving hospital operations

M. Ghosh & D. K. Sobek II, Journal of Health Organization and Management, 2015

<https://doi.org/10.1108/JHOM-09-2013-0191>

What is a Pareto Chart?

ASQ, 2026

<https://asq.org/quality-resources/pareto>

Kinds of accident statistics in Great Britain, 2025

Health and Safety Executive, 2025

<https://www.hse.gov.uk/statistics/causinj/index.htm>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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